Reproducibility Report: Coppock and Green (2015)

Assessing the Correspondence between Experimental Results Obtained in the Lab and Field: A Review of Recent Social Science Research

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Archived at Zenodo: doi:10.5281/zenodo.24168336. This DOI always resolves to the latest release of the maintained version.

Drafted by Claude Opus 5 under the supervision of Alex Coppock.

This repository holds the actively maintained replication code for Coppock and Green (2015), together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

Article	10.1017/psrm.2014.10
Replication archive	10.7910/DVN/25543
Errata	`coppock_green_2015_errata.pdf`, built from `errata.qmd`

The data are not redistributed here. The deposit is a single 52 KB RAR container holding 16 files, and it lives at Harvard Dataverse, which is the only copy this repository points at. `download_original.R` fetches it, verifies it, unpacks it and verifies each of the 16 members; `original_manifest.csv` pins the container and `original_members_manifest.csv` pins its contents, so the exact bytes this code was written against are recorded in version control even though the bytes themselves are not.

Repository layout. `maintained/` is the maintained rewrite: one script per published table or figure, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every published number to the code that produces it: `published_claims.csv` is the extraction of every numeric token in the article and its appendix, and `build_ground_truth.R` assembles the comparison table by reading every rewrite value back out of `maintained/output/`, so it cannot drift from the scripts. `maintained/in_text_claims.R` is a second, independent instrument that recomputes each of those numbers by its own path and prints it beside the sentence that states it; the build compares the two and stops if they disagree. `errata.qmd` builds the note listing the article's own misstated claims. `original/` and `original_extracted/` are created by the download script and are deliberately absent from the repository. This file is the reproducibility report, also available as a PDF in `report/`.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains. See `LICENSE`.

To reproduce. Clone or download the repository, open `coppock_green_2015.Rproj`, and run:

```
source("run_all.R")
```

That fetches the deposit, verifies it, produces every table and figure into `maintained/output/`, rebuilds the ground truth from those outputs, and finishes by printing the in-text claims audit. Required packages: tidyverse, estimatr, modelsummary, patchwork, knitr, kableExtra, archive, here. Paths resolve through `here`, so nothing depends on the working directory. The full run takes about 40 seconds, almost all of it in the two scripts that enumerate subsets of the 21 lab-field pairs: `figure_3_robustness.R` evaluates the rank-order correlation across 555,790 subsets, and `text_sampler_reproducibility.R` runs the same enumeration twice, once under each of R's two samplers. A successful run overwrites `maintained/output/`, which is committed: **`git diff` on that folder is the reproduction check.** Expect the CSV, TeX and PNG output to come back byte-identical and the three PDF figures to differ, because a PDF records the time it was written.

1 Summary

Two questions, answered before the detail.

1.1 Does the deposited archive run?

Yes, after one edit and two installations. Five of the six analysis scripts execute without error on a current R installation. The sixth, `Coppock Green Lab Field Appendix Table Code.R`, calls `setwd("")` on its seventh line and stops there; commenting that line out is the whole fix. Three of the scripts call `gplots` and `calibrate`, which have to be installed but do install from CRAN today, at versions 3.3.0 and 1.7.7.

Running is also reproducing, with one qualification. Of the five files the archive both writes and deposits, four come back unchanged: `meanstable.csv` character for character apart from line endings, `setoftwelve.rdata` to within 5e-16, and the two PDF figures with every mark in the same place. The fifth, `rankorderrobustness.pdf`, cannot come back unchanged, because footnote 6 samples 50,000 of the larger combination sets at random and R replaced the sampler behind `sample()` in version 3.6.0. That is a property of the environment rather than a defect in the deposit, and the section on Figure 3 below measures how much it moves.

Of the 33 published quantities an archive script can be checked against, 31 match the article to the precision the article prints. The 2 that do not are both the article's, not the code's: footnote 5 truncates 0.6995778 to 0.699 where the third decimal rounds to 0.700, and the text's assertion that the median correlation "stays constant across all permutations" is not what the archive's own procedure produces.

The one thing the archive cannot do is defend itself. Four of its six scripts write their outputs into the directory they are run from, and five of those outputs are themselves deposited files, so running the archive in place overwrites part of the deposit with the run's own output. That is why `download_original.R` unpacks into `original_extracted/` and leaves `original/` holding the container alone, and why the checks in this report were run in a scratch copy.

1.2 Does the maintained rewrite reproduce the paper?

Yes for the headline result, and it corrects three errors along the way. 45 of the 57 verifiable claims match the published values to reported precision, including the Spearman rank-order correlation of 0.73 that the abstract leads with. Of the 12 that do not, 4 differ because the rewrite corrects an error in the deposited code rather than because it fails to reproduce one; the Errata section below sets out all three corrections. The remaining 8 are sentences the article's own data do not support, and they are collected in `coppock_green_2015_errata.pdf` at the root of this repository, which quotes each published sentence and prints the corrected one with its values computed at render time.

The largest of the three is the disattenuated correlation, which the paper reports as 0.80 and the rewrite puts at 0.73. Footnote 7 defines the reliability of a treatment effect as the *square* of the correlation between two simulated draws, and the archive implements that definition faithfully. Spearman's correction for attenuation takes the reliability to be the correlation itself, so squaring it first inflates the corrected estimate. Correcting the definition, and two indexing errors that shift the raw correlation from 0.64 to 0.65, gives 0.73. The published conclusion is unaffected in direction: the disattenuated correlation still exceeds the observed one, and lab-field correspondence is still strong.

2 Paper overview

Citation: Coppock, A. and Green, D. P. (2015). "Assessing the correspondence between experimental results obtained in the lab and field: A review of recent social science research." *Political Science Research and Methods*, 3(1), 113-131. DOI: 10.1017/psrm.2014.10

Summary: The article reviews the literature comparing experimental results obtained in laboratory and field settings, and reanalyzes twelve pairs of studies conducted in parallel across the two. Because the outcome scales are incomparable across pairs (mazes solved against footrace times, hypothetical donations against real ones), correspondence is assessed by rank-order correlation rather than by comparing effect sizes directly. Each study's treatment effects are standardized by the control group standard deviation, which yields 21 lab-field pairs of standardized effects across the twelve studies. The Spearman rank-order correlation of those 21 pairs is 0.73, and a leave-out and duplicate-in robustness exercise shows the interquartile range staying within a band of about 0.6 to 0.8 under every way of dropping or duplicating up

to ten pairs. The article closes by noting that the set of available lab-field comparisons is itself an ad hoc and possibly selected one, and calls for comparisons designed rather than found.

3 Original archive reproducibility

The deposit is one RAR container with 16 files: six analysis scripts, one duplicate script, one data file, a readme, five deposited outputs, and two files the author's operating system and R session left behind (`.DS_Store` and a 26 KB `.Rhistory`).

Table 1: Original archive reproducibility, checked against R 4.6.0.

Script	Status on current R	Resolution
Coppock Green Lab Field Tables 1 and 2 Code.R	Clean	No changes required
Coppock Green Lab Field Figure 1 Code.R	Clean once gplots and calibrate are installed	install.packages(c('gplots', 'calibrate'))
Coppock Green Lab Field Figure 2 Code.R	Clean once gplots and calibrate are installed	install.packages(c('gplots', 'calibrate'))
Coppock Green Lab Field Figure 3 Code.R	Clean once gplots and calibrate are installed	install.packages(c('gplots', 'calibrate'))
Coppock Green Lab Field Appendix Table Code.R	Stops on line 7	Comment out setwd("")
Coppock Green Lab Field Disattenuation Code.R	Clean	No changes required
tables1and2.r	Undocumented duplicate of the Tables 1 and 2 script, minus its header	None needed; not run

`setwd("")` is a placeholder the author left uncommented in one script out of six. The same line is present but commented out in four of the other five, which is what makes the omission legible as an oversight rather than a design. `gplots` and `calibrate` are both still on CRAN and both install without incident, at 3.3.0 and 1.7.7; a report of missing packages is a report about a machine, not about availability.

Figure 3's script is a small curiosity: it calls `library(gplots)` at line 74, well after the loop that computes every number the figure reports. On a machine without `gplots` it therefore prints all of its results and then fails at the plotting stage, which is a more forgiving failure than it looks.

3.1 What the archive regenerates

Four scripts write files, and five of the files they write are deposited alongside them. Re-running the archive from a scratch copy regenerates four of the five exactly and the fifth up to the random sampling the article documents.

Table 2: What the deposited code produces today, compared with the outputs deposited beside it.

Deposited output	Produced by	Regenerated result
meanstable.csv	Appendix Table Code.R	Identical apart from line endings
setoftwelve.rdata	Figure 2 Code.R	Identical to within 5e-16
tiles_landscape.pdf	Figure 1 Code.R	Every mark in the same place; the file differs only in how the current PDF device encodes plotting symbols
standardized.pdf	Figure 2 Code.R	Every mark in the same place; the file differs only in how the current PDF device encodes plotting symbols
rankorderrobustness.pdf	Figure 3 Code.R	Coordinates differ; Figure 3 samples 50,000 of the larger combination sets and R 3.6.0 changed the sampler

The two PDF results deserve the qualifier they carry. Neither file is byte-identical to its deposited twin, and decompressing the content streams shows that every difference is one of two changes in R's PDF device rather than a change in the plot. An open plotting symbol used to be written as a glyph from the symbol font and is now written as a Bézier path drawn at the same centre: 42 symbols in `standardized.pdf` and 147 in `tiles_landscape.pdf`. A closed polyline that used to end `l S` now ends `h S`, once in the first file and twelve times in the second. Rendering both versions of each figure at 200 dpi and differencing them changes 0.02 per cent of the pixels in each, all of them on the anti-aliased edges of those redrawn symbols, and nothing anywhere else on either page.

That is a stronger result than it sounds. The archive's numbers are the paper's numbers, and the code that produced them still produces them.

3.2 Figure 3 and the sampler

`rankorderrobustness.pdf` is the exception, and the reason is in the article. Footnote 6 records that when more than 50,000 combinations are available, 50,000 of them are sampled at random. R 3.6.0 replaced the algorithm behind `sample()`, so code written in 2014 draws different subsets today at the same seed, and the archive's Figure 3 draws at ten of its twenty subset sizes.

`maintained/text_sampler_reproducibility.R` measures that directly. It runs the archive's own procedure twice on the archive's own deposited effects, once with `RNGkind(sample.kind = "Rounding")` to restore the sampler of the day and once under the current default, and compares the 100 summary values behind the figure.

Table 3: The archive's Figure 3 procedure run under the sampler of 2014 and under the current one.

Quantity	Value

Summary values compared	100
Values identical under the two samplers	74
Values differing by 0.005 or more	6
Largest absolute difference	0.0736

The five-number summary is barely moved: most values are identical and the largest change is 0.074, at the extremes where a single unlucky subset can be drawn or missed. The maintained rewrite keeps the current sampler, because it is the code we would write now. Where that moves a number, the difference belongs to R rather than to either analysis.

4 Errata in the deposited code

Three errors, all of them in the archive rather than introduced by it, and all of them corrected in the rewrite. None changes the article's conclusion. Sentences in the article that its own data do not support are a separate matter and are collected in `coppock_green_2015_errata.pdf`; the first of the three below is the one place the two overlap, because the article's footnote 7 states in words what the deposited script does in code.

4.1 The reliability is squared

`Coppock Green Lab Field Disattenuation Code.R` estimates the reliability of the lab and field treatment effects by simulation: for each of the 21 pairs it draws two treatment effects from the normal distribution implied by the estimate and its standard error, correlates the first draws against the second, and averages over 10,000 simulations. The line that stores the result squares it:

```
lab.relys[j] <- cor(draw1.lab, draw2.lab)^2
```

Footnote 7 of the article says the same thing in words: "The estimated reliability is the square of the correlation between two treatment effect draws." Code and text agree, and both are wrong. Under classical test theory the reliability of a measure is the correlation between two parallel measurements of it, and Spearman's correction for attenuation divides the observed correlation by the geometric mean of the two reliabilities. Squaring the correlations before averaging them makes the denominator too small and the corrected estimate too large. The error was identified by David Broockman (UC Berkeley) in January 2024. No corrigendum has been published, so this repository is where the correction is recorded. Broockman's own figures, computed without the two indexing corrections described next, put the field reliability at 0.91 and the disattenuated correlation at 0.72; the rewrite's 0.90 and 0.73 differ from them for that reason.

4.2 The female treatment effect is standardized by a male treatment group

`Coppock Green Lab Field Figure 2 Code.R` computes two standardized treatment effects for the Gneezy and Rustichini (2004) comparison, one for men and one for women. The male effect contrasts row 3 (Competition, M) against row 1 (No Competition, M); the female effect contrasts row 4 (Competition, F) against row 2 (No Competition, F). Both are then standardized:

```
s.gender.lab.tes <- gender.lab.tes / gender.lab.sds[c(1, 3)]
```

The first divisor is right: row 1 is the male control group. The second is not: row 3 is the male *treatment* group, where the female comparison's control group is row 2. Equation 1 of the article is explicit that the divisor is the control group standard deviation. The same `c(1, 3)` index appears eight times in the script, four times for the lab quantities and four times for the field, so the estimate, both interval bounds and the standard error are all affected in both settings. The rewrite standardizes the female effect by row 2, which is what the equation says and what every other pair in the paper does.

4.3 The List (2006) field standard error is the lab standard error

The same script assembles the List (2006) results:

```
list.field.tes.ses <- c(list.lab.2.1[4])
```

Every neighboring line reads its value from `list.field.2.1`; this one reads from `list.lab.2.1`. The standard error does not enter Figure 2, whose intervals come from separately stored bounds, so the published figure is unaffected. It does enter the reliability simulation, which draws from the standard errors, so it moves the field reliability and through it the disattenuated correlation.

4.4 What the three corrections do

Table 4: The disattenuation calculation under the published definition and under the corrected one. The middle column keeps footnote 7's squared reliability and fixes only the two indexing errors; the right column is what the rewrite reports.

Quantity	Published	Indexing fixed only	Indexing and reliability fixed
Raw Pearson correlation, 21 pairs	0.64	0.65	0.65
Lab reliability	0.78	0.78	0.88
Field reliability	0.82	0.82	0.90

Disattenuated correlation	0.80	0.81	0.73

The two indexing errors alone move the disattenuated correlation from 0.80 to 0.81. Correcting the reliability definition as well moves it to 0.73. Both reliabilities rise sharply, from 0.78 to 0.88 in the lab and from 0.82 to 0.90 in the field, because a correlation near 0.9 is much larger than its own square.

The rank-order correlation the paper leads with is untouched: it moves from 0.7275 to 0.7314, which rounds to 0.73 either way. The two bounds in footnote 5, which report how far the pooled correlation moves as the single included Jerit, Barabas and Clifford pair varies across the 17 candidates, shift from 0.699 and 0.762 to 0.705 and 0.767. Since the indexing correction also reorders the 17 candidates, the median case is now candidate 12 rather than candidate 3.

5 Number-by-number comparison

Table 5: Ground truth: published value against the value the deposited scripts produce on current R. A blank Archive cell means no deposited script computes the quantity; a blank Paper cell means the article and its available appendix do not state it.

Location	Quantity	Paper	Archive	Match
Table 1, p. 120	Pair 1, lab ATE	2.0		
Table 1, p. 120	Pair 1, field ATE	0.5		
Table 1, p. 120	Pair 2, lab ATE	5.0		
Table 1, p. 120	Pair 2, field ATE	4.0		
Table 1, p. 120	Pair 3, lab ATE	3.0		
Table 1, p. 120	Pair 3, field ATE	3.0		
Table 1, p. 120	Pair 4, lab ATE	0.0		
Table 1, p. 120	Pair 4, field ATE	3.0		
Table 1, p. 120	Pair 5, lab ATE	2.0		
Table 1, p. 120	Pair 5, field ATE	-1.5		
Table 1, p. 120	Pair 6, lab ATE	0.5		
Table 1, p. 120	Pair 6, field ATE	4.0		
Table 1, p. 120	Pair 7, lab ATE	-2.0		
Table 1, p. 120	Pair 7, field ATE	1.5		
Table 1, p. 120	Pair 8, lab ATE	-2.0		
Table 1, p. 120	Pair 8, field ATE	2.0		
Table 2, p. 120	Subjects (A)	2.00	2	1
Table 2, p. 120	Treatments (T1)	1.00	1	1
Table 2, p. 120	Outcomes (Effort)	1.00	1	1
Table 2, p. 120	Contexts (Field)	1.00	1	1
Table 2, p. 120	Constant	-0.94	-0.9375	1
Text, p. 120	Units A-type subjects add, quoted into prose	2	2	1
Text, p. 120	Units treatment T1 adds, quoted into prose	1	1	1
Text, p. 120	Units the effort outcome adds, quoted into prose	1	1	1
Text, p. 120	Units the field context adds, quoted into prose	1	1	1
Table 3, p. 125	Parallel lab-field study pairs	12	12	1
Text, p. 120	Correlation across the eight hypothetical pairs	0.14		
Text, p. 121	Correlation across the eight hypothetical pairs, restated a page later	0.14		
Text, p. 121	Correlation when Pairs 1 and 5 go unreported	0.76		

Table 5: Ground truth: published value against the value the deposited scripts produce on current R. A blank Archive cell means no deposited script computes the quantity; a blank Paper cell means the article and its available appendix do not state it. *(continued)*

Location	Quantity	Paper	Archive	Match
Figure 1, p. 124	Jerit panel, Pearson correlation	0.21	0.211055	1
Figure 1, p. 124	Jerit panel, Spearman correlation	0.26	0.259963	1
Figure 2, p. 126	Pooled treatment effect pairs	21	21	1
Figure 2, p. 126	Rank-order correlation printed on the figure	0.73	0.727509	1
Results, p. 125	Rank-order correlation in the sentence describing Figure 2	0.73	0.727509	1
Abstract, p. 113	Rank-order correlation	0.73	0.727509	1
Introduction, p. 116	Rank-order correlation	0.73	0.727509	1
Footnote 5, p. 124	Jerit candidate pairs	17	17	1
Footnote 5, p. 124	Lowest pooled rho across the candidates	0.699	0.699578	0
Footnote 5, p. 124	Highest pooled rho across the candidates	0.762	0.761936	1
Footnote 5, p. 125	Pooled rho including all 17 Jerit pairs	0.56		
Footnote 6, p. 125	Combinations sampled where more than 50,000 exist	50000	50000	1
Footnote 6, p. 125	Smallest number of dropped pairs at which the cap binds	7		
Figure 3	Lowest 25th percentile across the combination sizes		0.633259	
Figure 3	Highest 75th percentile across the combination sizes		0.790909	
Results, p. 126	Interquartile range stays within 0.6 to 0.8 across every combination size	1	1	1
Results, p. 126	Minimum correlation crosses zero when any ten pairs are dropped	1	1	1
Results, p. 126	Highest correlation when any ten pairs are dropped	1	0.972727	
Results, pp. 125-126	Distinct values of the median correlation, to two decimal places	1	2	0
Figure 3 note, p. 127	Every available combination is evaluated	1		
Figure 3	Largest change in a plotted summary value under the R 3.6 sampler		0.0735553	
Discussion, p. 127	Median lab sample size per treatment condition	40		
Discussion, p. 127	Median field sample size per treatment condition	37.5		
Discussion, p. 128	Raw Pearson correlation across the 21 pairs	0.64	0.642814	1
Discussion, p. 128	Lab reliability	0.78	0.78334	1
Discussion, p. 128	Field reliability	0.82	0.823976	1
Discussion, p. 128	Raw Pearson correlation restated inside the disattenuation formula	0.64	0.642814	1
Discussion, p. 128	Lab reliability restated inside the disattenuation formula	0.78	0.78334	1
Discussion, p. 128	Field reliability restated inside the disattenuation formula	0.82	0.823976	1
Discussion, p. 128	Disattenuated correlation	0.80	0.800116	1
Footnote 7	Lab reliability under the published definition	0.78	0.78334	1
Footnote 7	Field reliability under the published definition	0.82	0.823976	1
Footnote 7	Disattenuated correlation under the published definition		0.800116	
Online Appendix Table 1	Treatment condition rows		32	
Online Appendix Table 1	Cells agreeing with the deposited meanstable.csv		192	
Online Appendix Table 2	Descriptions of the twelve study pairs			

Of the 65 recorded claims, 33 can be compared against what a deposited script prints, and 31 of those match the published value. 2 fail, and in both the article is the party at fault. The first is footnote 5, which gives the range of pooled correlations as 0.699 to 0.762 where the script prints 0.6995778, a value that rounds up to 0.700 at the three decimals the footnote uses. The second is the sentence describing Figure 3, which says the median correlation "stays

constant across all permutations", where the archive's own procedure gives 2 distinct values at two decimal places, running from 0.720 to 0.729. Neither changes anything the paper concludes, and neither is a failure of the code.

Beyond the stipulated entries of Table 1, 8 quantities the article states are ones no deposited script computes: the two correlations quoted from the Table 1 illustration and one of them restated a page later, the pooled correlation when all 17 Jerit pairs are included, the two median sample sizes in the Discussion, and the two counts behind footnote 6 and the note to Figure 3. All are recoverable from the deposited data; the rewrite adds `text_hypothetical_correlations.R`, `text_jerit_sensitivity.R` and `text_median_sample_sizes.R`, and records the combination counts behind Figure 3 alongside the correlations it plots.

5.1 Every number in the article, and a second instrument

The table above starts from the pipeline and asks what each output corresponds to in the article. That question cannot find a claim the pipeline never reaches, so the repository also works in the other direction. `ground_truth/published_claims.csv` is an exhaustive extraction: every numeric token in the body, the footnotes, the tables, the figures and the published appendix, read off the typeset page rather than searched for by pattern, together with a companion pass for numbers the article spells out in words, which no token scan sees. Each of the 134 resulting claims is classified by hand.

Table 6: The extraction of every number the article states, by kind of claim. Pipeline and descriptive claims must be reproduced from the data; definitional and structural ones are checked wherever the pipeline can reach them and against the article's own methods otherwise; transcribed ones belong to the papers they were copied from.

Type	Verified at the point of use	Checked by a block
definitional	5	32
descriptive	0	6
pipeline	0	30
structural	11	47
transcribed	3	0

`maintained/in_text_claims.R` carries one block per checkable claim. Each block quotes the article's sentence verbatim, then recomputes the number from `maintained/output/` and prints it at the precision the page uses. It never reads the ground truth, so the two instruments reach the same 36 pipeline and descriptive quantities by separate paths, and `build_ground_truth.R` compares them value by value and halts if any pair disagrees. It also halts if the claims script prints a claim the extraction does not list, if the extraction lists a claim the script does not print, or if a pipeline or descriptive claim has no row in the ground truth. All 115 claims that require a block print one.

Working from the article rather than from the outputs is what turned up footnote 6 and the note to Figure 3, neither of which corresponds to any pipeline output and neither of which the number-by-number table could have posed as a question. It also gives the figures coverage they would not otherwise have: Figure 1 prints no estimates on its face, so what each of its twelve panels asserts is the number of points it draws, and all twelve counts are checked against the published page.

7 rows have no published value to compare against. 2 of them concern Online Appendix Table 1, the table of group means behind every panel of Figure 1. The appendix published with the article is the bibliography of the studies considered for inclusion, 80 entries against a Method section that says approximately 80, and neither of the two online appendix tables the text refers to is in it, so the 192 cells of that table cannot be checked against a published source. They are checked instead against `meanstable.csv`, the deposit's own copy of the table, and all 192 agree exactly at the published rounding. One row is Online Appendix Table 2, which is prose. The other 3 are Figure 3, which plots points and prints no numbers.

6 Maintained rewrite

The rewrite lives in `maintained/`: a shared `helpers.R` and fourteen scripts covering one cleaning step, three published tables, three published figures, five sets of in-text quantities, the sampler comparison behind the Figure 3 section above, and the claims audit described earlier. It is a translation, not a reanalysis, except for the three errata above: every estimator, standardization and sample definition is the one the paper used.

6.1 Architecture

`clean_labfield.R` is the only script that touches the deposit. It reads the twelve data frames in `labfieldframes.rdata`, tidies the group means into one long table, and computes the 21 standardized treatment effects from the group means, standard deviations and sizes, following Equations 1 and 2 of the article. Everything downstream reads its two outputs.

The original code carries the study prefix in every column name, so `erev.frame` has an `erev.lab.means` column and `gneezy.frame` has a `gneezy.lab.means` column, and it reaches them by calling `attach()` on all twelve frames in turn. The rewrite strips the prefixes once, in `normalize_frame()`, and drives everything from an explicit table of study specifications: which frame, which column holds the condition labels, which rows form each treatment-control contrast, and which row's standard deviation standardizes it. The three errata are one-character changes in that table rather than edits scattered through 431 lines.

Armantier and Boly's outcome is binary, so its effects are differences in proportions rather than differences in means, and the archive handles it by writing out eight vectors of zeros and ones by hand. Those vectors are not free-floating constants: each is exactly determined by the

group size and group mean in `boly.frame`, so the rewrite rebuilds them with `binary_group()` and no count is typed into the code.

The hypothetical treatment effects of Table 1 are the one set of numbers the rewrite does type in, because they are the paper's stipulated premises rather than estimates of anything. They are written once, in `table_1_hypothetical_ates.R`, and both the Table 2 regression and the two correlations quoted in the text read them back from that script's committed output rather than repeating them. The archive types them twice, once in each of its two copies of the Tables 1 and 2 script.

6.2 Deprecated patterns replaced

Table 7: Deprecated patterns and their replacements in the maintained rewrite.

Original pattern	Replacement
‘rm(list = ls())‘	(omitted)
‘setwd("")‘	‘here::here()‘
‘attach(frame)‘, twelve times per script	‘normalize_frame()‘ and explicit ‘$‘ access
‘gplots::plotCI(err = "x")‘ / ‘plotCI(err = "y")‘	‘geom_linerange()‘
‘calibrate::textxy()‘	‘geom_text()‘
‘xtable‘ + ‘print.xtable‘	‘knitr::kable()‘, plus ‘write_csv()‘ to ‘output/‘
‘lm()‘	‘estimatr::lm_robust()‘, and ‘modelsummary()‘ for the table
‘pdf()‘ / ‘dev.off()‘	‘ggsave()‘
‘par(mfrow = c(3, 4))‘	‘patchwork::wrap_plots()‘
‘data.frame()‘	‘tibble()‘
‘for‘ loops over preallocated ‘rep(NA, n)‘ vectors	‘map()‘ returning tibbles, ‘list_rbind()‘
hardcoded figure legends	values computed and formatted at plot time

Two of these deserve a word. The archive's Figure 1 legend reads `legend(..., legend = c("Pearson Correlation = 0.21", "Spearman Correlation = 0.26"))`, with the numbers typed in, ten lines below the `cor()` calls that compute them. They happen to be right. The rewrite formats the computed values into the annotation, so the figure cannot drift from the data behind it. Second, the rewrite's Figure 1 keeps each panel's own axis labels and limits, which is why it assembles twelve separate plots with `patchwork` rather than faceting: `facet_wrap()` would force one pair of axis labels on all twelve, and the panels measure kilos of oranges, maze completions, restaurant bills and bribe-taking rates.

7 Figures

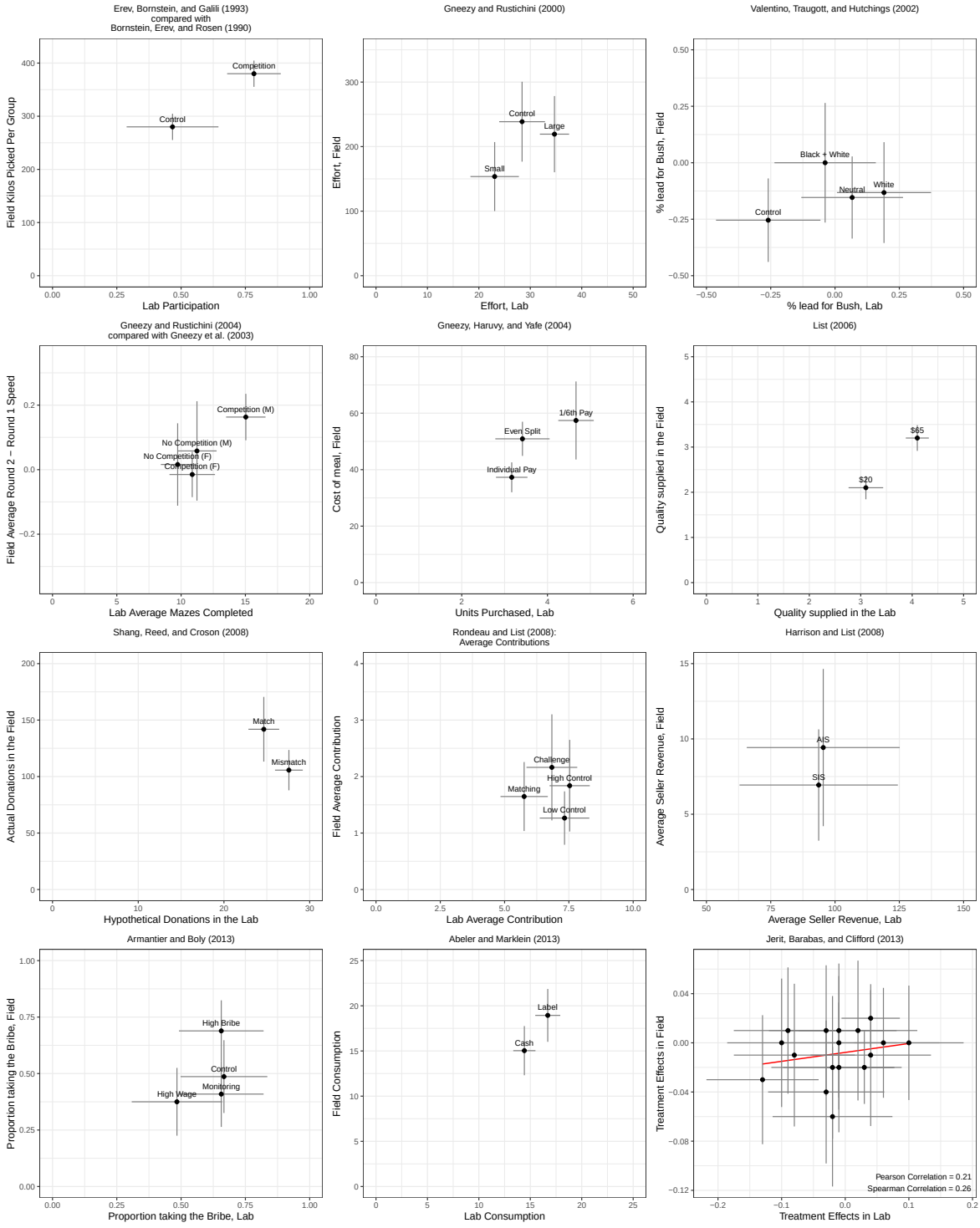


Figure 1: Figure 1 as reproduced by the maintained rewrite. Lab results on the horizontal axis, field results on the vertical, with 95 percent confidence intervals. Eleven panels show treatment group means; the Jerit, Barabas and Clifford panel shows treatment effects.

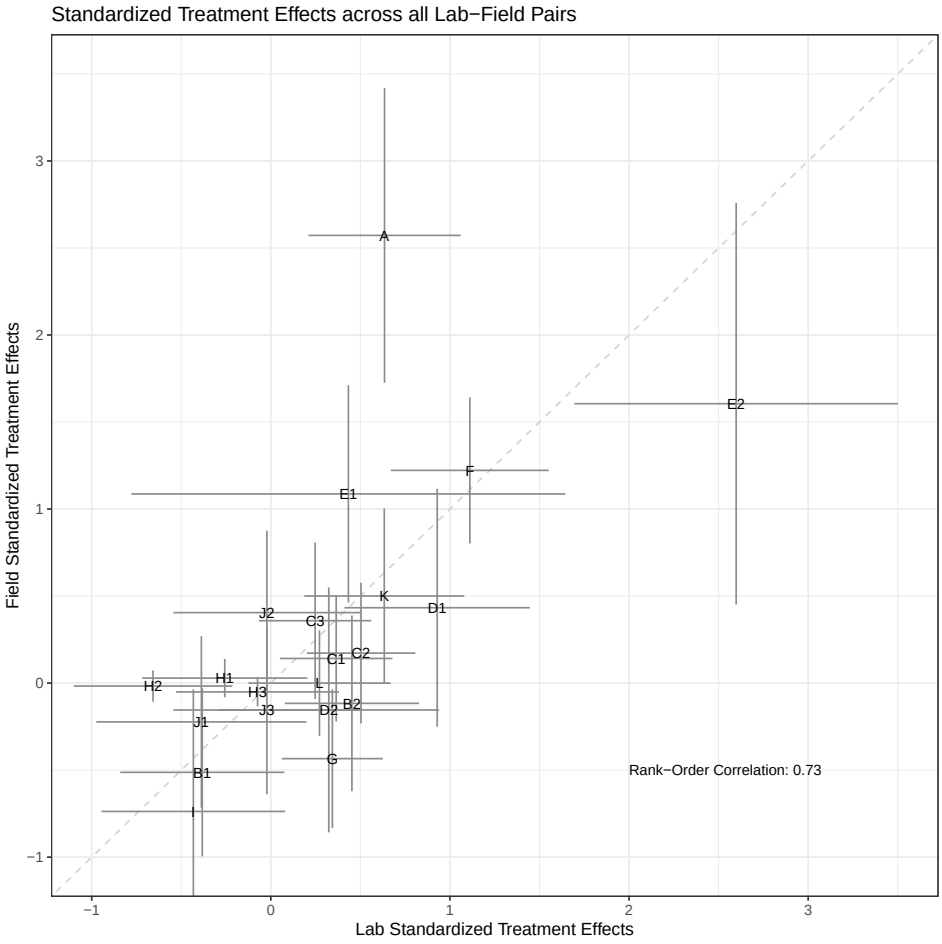


Figure 2: Figure 2 as reproduced by the maintained rewrite. The 21 standardized lab-field pairs, labelled as in the article, with the rank-order correlation that summarizes them.

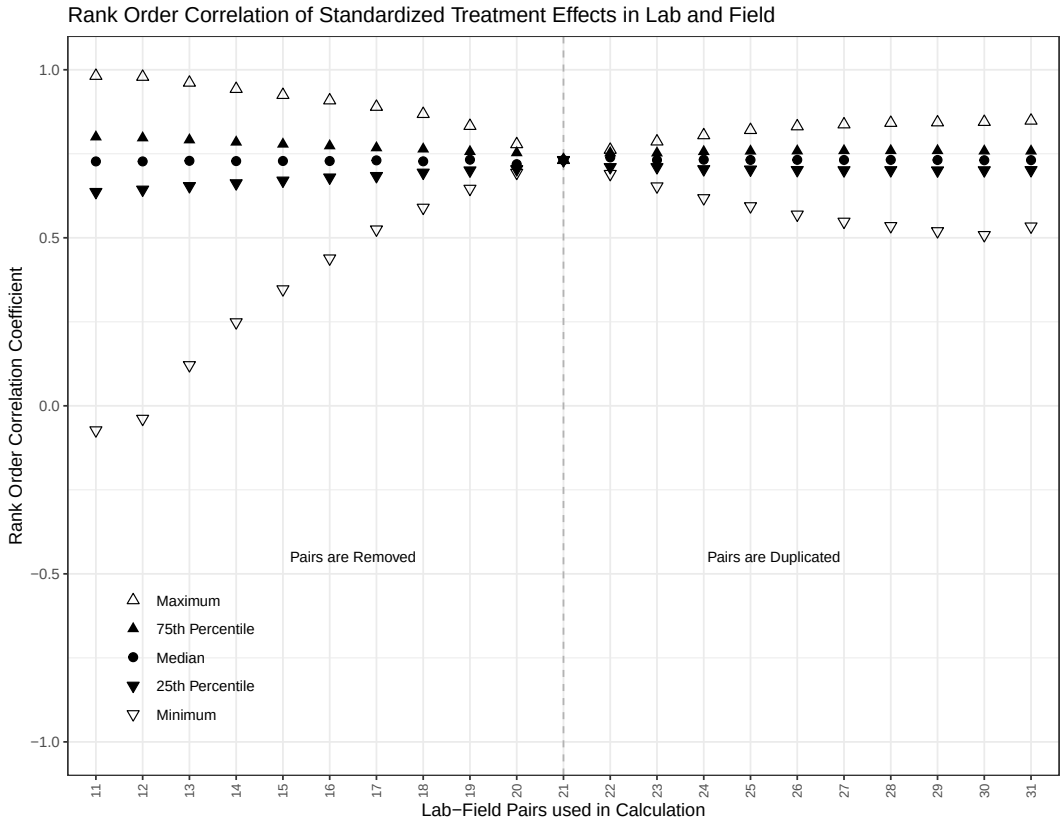


Figure 3: Figure 3 as reproduced by the maintained rewrite. The five-number summary of the rank-order correlation under every way of removing 1 to 10 of the 21 pairs and of duplicating 1 to 10 of them.

Figure 3 is the one result that cannot be reproduced exactly, for the two reasons set out above: footnote 6's random sampling of 50,000 combinations, and the change of sampler in R 3.6.0. The archive and the rewrite also consume the random stream in a different order, so they would draw different subsets even under one sampler. What the figure claims is unaffected. Across every combination size the archive's lowest 25th percentile is 0.633 and its highest 75th percentile is 0.791; in the rewrite they are 0.636 and 0.800. The article's claim is that the interquartile range stays within a band of about 0.6 to 0.8, and both runs put it there. Its other claim about the figure, that the minimum correlation crosses zero once any ten pairs are dropped, also holds in both.

The one thing the figure's accompanying sentence gets wrong is the median. The text says it "stays constant across all permutations". It does not, in either run: the rewrite's median takes 3 distinct values at two decimal places, running from 0.719 to 0.739. The variation is small enough that the substantive point survives, but constant it is not.

The reliability simulation is a different matter, and there the rewrite matches the archive's random draw exactly rather than approximately. The archive draws in the order lab, lab, field, field for each pair before moving to the next pair and then to the next simulation; `text_disattenuation.R` hands `rnorm()` its means in that same order, so the two runs consume identical streams and the difference between their reliabilities is entirely attributable to the corrections rather than partly to Monte Carlo noise.

8 Maintained rewrite verification

Table 8: Maintained rewrite verification: published value against rewrite output.

Location	Quantity	Paper	Rewrite	Match
Table 1, p. 120	Pair 1, lab ATE	2.0	2	1
Table 1, p. 120	Pair 1, field ATE	0.5	0.5	1
Table 1, p. 120	Pair 2, lab ATE	5.0	5	1
Table 1, p. 120	Pair 2, field ATE	4.0	4	1
Table 1, p. 120	Pair 3, lab ATE	3.0	3	1
Table 1, p. 120	Pair 3, field ATE	3.0	3	1
Table 1, p. 120	Pair 4, lab ATE	0.0	0	1
Table 1, p. 120	Pair 4, field ATE	3.0	3	1
Table 1, p. 120	Pair 5, lab ATE	2.0	2	1
Table 1, p. 120	Pair 5, field ATE	-1.5	-1.5	1
Table 1, p. 120	Pair 6, lab ATE	0.5	0.5	1
Table 1, p. 120	Pair 6, field ATE	4.0	4	1
Table 1, p. 120	Pair 7, lab ATE	-2.0	-2	1
Table 1, p. 120	Pair 7, field ATE	1.5	1.5	1
Table 1, p. 120	Pair 8, lab ATE	-2.0	-2	1
Table 1, p. 120	Pair 8, field ATE	2.0	2	1
Table 2, p. 120	Subjects (A)	2.00	2	1
Table 2, p. 120	Treatments (T1)	1.00	1	1
Table 2, p. 120	Outcomes (Effort)	1.00	1	1
Table 2, p. 120	Contexts (Field)	1.00	1	1
Table 2, p. 120	Constant	-0.94	-0.9375	1
Text, p. 120	Units A-type subjects add, quoted into prose	2	2	1
Text, p. 120	Units treatment T1 adds, quoted into prose	1	1	1
Text, p. 120	Units the effort outcome adds, quoted into prose	1	1	1
Text, p. 120	Units the field context adds, quoted into prose	1	1	1
Table 3, p. 125	Parallel lab-field study pairs	12	12	1
Text, p. 120	Correlation across the eight hypothetical pairs	0.14	0.139999	1
Text, p. 121	Correlation across the eight hypothetical pairs, restated a page later	0.14	0.139999	1
Text, p. 121	Correlation when Pairs 1 and 5 go unreported	0.76	0.764266	1
Figure 1, p. 124	Jerit panel, Pearson correlation	0.21	0.211055	1
Figure 1, p. 124	Jerit panel, Spearman correlation	0.26	0.259963	1
Figure 2, p. 126	Pooled treatment effect pairs	21	21	1
Figure 2, p. 126	Rank-order correlation printed on the figure	0.73	0.731406	1
Results, p. 125	Rank-order correlation in the sentence describing Figure 2	0.73	0.731406	1
Abstract, p. 113	Rank-order correlation	0.73	0.731406	1
Introduction, p. 116	Rank-order correlation	0.73	0.731406	1
Footnote 5, p. 124	Jerit candidate pairs	17	17	1
Footnote 5, p. 124	Lowest pooled rho across the candidates	0.699	0.705424	0
Footnote 5, p. 124	Highest pooled rho across the candidates	0.762	0.767132	0
Footnote 5, p. 125	Pooled rho including all 17 Jerit pairs	0.56	0.558066	1
Footnote 6, p. 125	Combinations sampled where more than 50,000 exist	50000	50000	1
Footnote 6, p. 125	Smallest number of dropped pairs at which the cap binds	7	6	0

Table 8: Maintained rewrite verification: published value against rewrite output. *(continued)*

Location	Quantity	Paper	Rewrite	Match
Figure 3	Lowest 25th percentile across the combination sizes		0.636364	
Figure 3	Highest 75th percentile across the combination sizes		0.8	
Results, p. 126	Interquartile range stays within 0.6 to 0.8 across every combination size	1	1	1
Results, p. 126	Minimum correlation crosses zero when any ten pairs are dropped	1	1	1
Results, p. 126	Highest correlation when any ten pairs are dropped	1	0.981818	
Results, pp. 125-126	Distinct values of the median correlation, to two decimal places	1	3	0
Figure 3 note, p. 127	Every available combination is evaluated	1	0	0
Figure 3	Largest change in a plotted summary value under the R 3.6 sampler			
Discussion, p. 127	Median lab sample size per treatment condition	40	40	1
Discussion, p. 127	Median field sample size per treatment condition	37.5	37.5	1
Discussion, p. 128	Raw Pearson correlation across the 21 pairs	0.64	0.649911	0
Discussion, p. 128	Lab reliability	0.78	0.882705	0
Discussion, p. 128	Field reliability	0.82	0.903386	0
Discussion, p. 128	Raw Pearson correlation restated inside the disattenuation formula	0.64	0.649911	0
Discussion, p. 128	Lab reliability restated inside the disattenuation formula	0.78	0.882705	0
Discussion, p. 128	Field reliability restated inside the disattenuation formula	0.82	0.903386	0
Discussion, p. 128	Disattenuated correlation	0.80	0.727795	0
Footnote 7	Lab reliability under the published definition	0.78	0.781773	1
Footnote 7	Field reliability under the published definition	0.82	0.817887	1
Footnote 7	Disattenuated correlation under the published definition		0.812768	
Online Appendix Table 1	Treatment condition rows		32	
Online Appendix Table 1	Cells agreeing with the deposited meanstable.csv		192	
Online Appendix Table 2	Descriptions of the twelve study pairs			

45 of the **57** verifiable claims match the published paper to reported precision. Of the **12** that do not, 4 are consequences of the three code errata and carry a note in the ground truth saying which; the other 8 are claims the article makes that neither the archive nor the rewrite supports. Every one of those rows carries a `defect_locus` recording where the fault lies, and none of them is the rewrite.

Determinism was verified by running `run_all.R` twice from a clean session and comparing every file it writes. All twenty CSV, TeX and PNG files in `maintained/output/`, and the ground truth CSV rebuilt from them, came back byte-identical; the three PDF figures differ, as they must, because a PDF records the time it was written, and they are identical once the embedded `CreationDate` and `ModDate` are stripped.

9 R environment

Item	Value
R version	4.6.2
Platform	aarch64-apple-darwin23

Date run	2026-11-12

Table 10: Package versions used for the run behind this report.

Package	Version
tidyverse	2.0.0
dplyr	1.2.1
ggplot2	4.0.3
purrr	1.2.2
estimatr	2.0.1
modelsummary	2.6.0
patchwork	1.3.2
knitr	1.52
kableExtra	1.4.1
archive	1.1.14
here	1.0.2